

MEDIATEK

MTK_Log之SOS shutdown日志落盘flash

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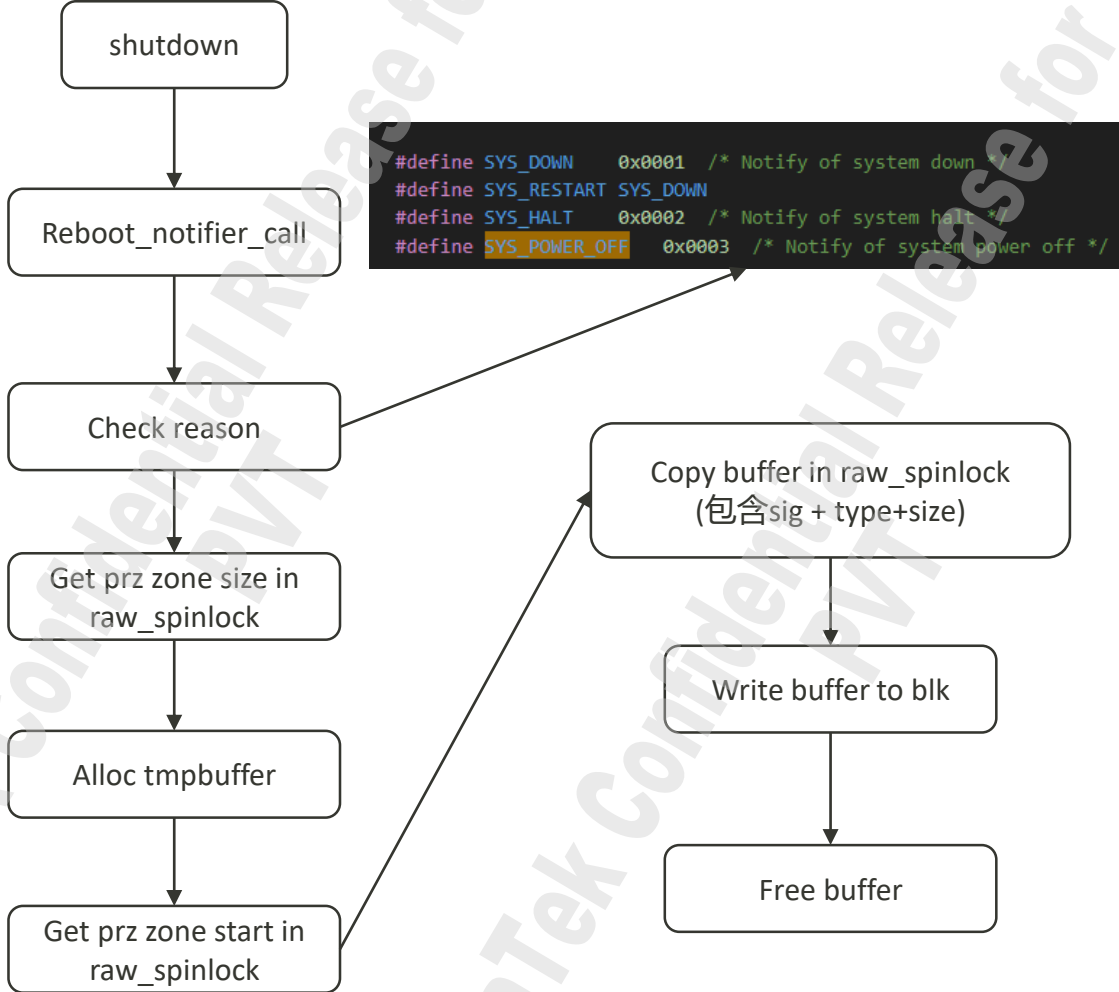
目的

- 在shutdown/reboot的过程中，Yocto端mobile_log_d进程会被systemd第二轮（发送sigterm）杀掉，后续log无法落盘；
- 现状：last_kmsg机制是将log通过ramoops写到DRAM中，而在shutdown最后阶段SOC会掉电，DRAM中的内容也会丢失；
- 新增：增加pstore后端为blk，在shutdown过程中，blk设备卸载前将现场的kmsg/pmsg存到pstore的裸分区中，blk数据掉电也能保持，在下一次开机将日志读取出来存储到文件系统。

方案流程

写入blk部分

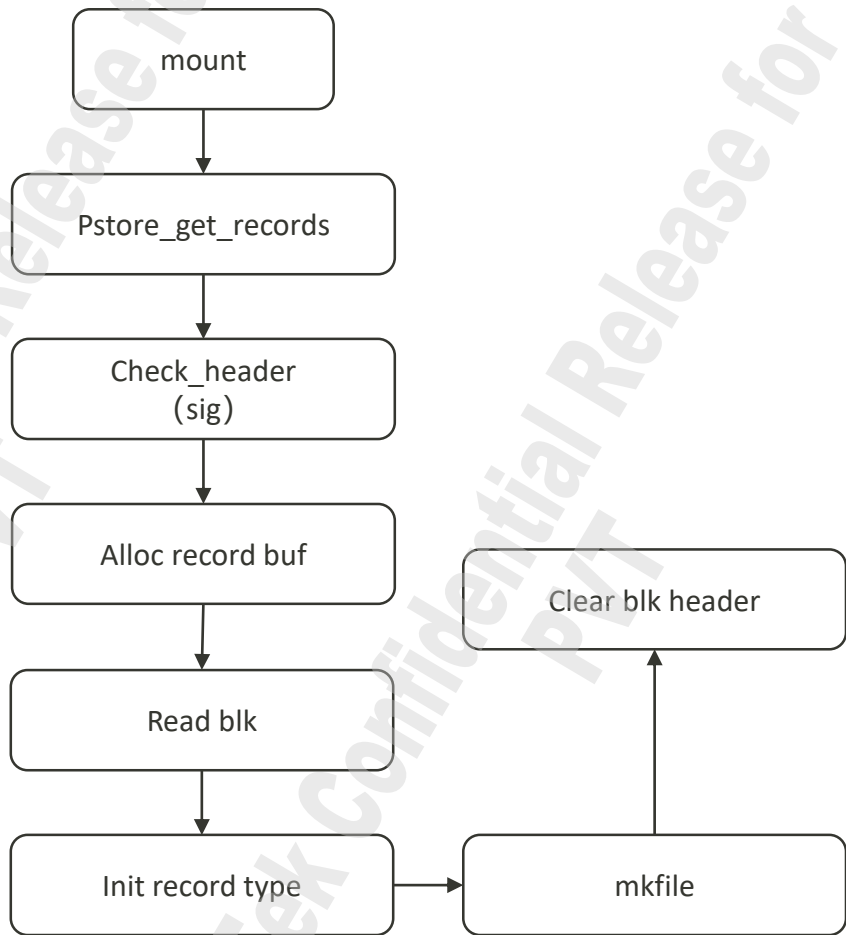
1. pstore 注册reboot_notifier
2. reboot_notifier回调函数判断reboot reason是否为SYS_POWER_OFF (目的: 只在shutdown做dump到blk)
3. 判断blk节点能否打开
4. copy prz到buffer中
5. 判断数据是否有效
6. 将buffer数据写入blk节点



方案流程

读取blk到文件系统

- 1.判断pstore blk节点能否打开
- 2.读取pstore blk中的header是否为PERSISTENT_RAMBLK_SIG
- 3.读取pstore blk中的buf
- 4.创建文件用来存buf内容
- 5.清除pstore blk中的header
- 6.将buf内容写入文件



注意事项

- blk 设备路径只能使用/dev/xxx, 请不要使用其他路径及软连接
 - 有可能在启动的时候软链接等路径没ready 导致生成日志失败
 - pstore blk设备路径通过config配置: `CONFIG_PSTORE_RAM_BLK_BLKDEV = "/dev/sdc18"`
 - lsblk 命令查看可用blk设备, 建议大于1M
- 正常启动后如果pstore 驱动正常读取日志后会对blk header 部分进行清零处理;
 - 目的: blk中的内容不清除会一直存在, 为防止每次上电都有日志混淆分析, 需要主动清除flag;
 - 每一次dump都会重新写入header, 读取完后清除header
- last_kmsg_blk 最大 256KB; last_journallog_blk 64KB
 - 取决于lk中reserve pstore memory的大小
 - from lk2 source: platform/mediatek/common/aee/include/platform/mboot_params.h
 - `+#define PSTORE_CONSOLE_SIZE (0x40000)`
 - `+#define PSTORE_PMSG_SIZE (0x10000)`
- APlog 文件夹中读取: 内核日志 last_kmsg_blk; 用户态日志 last_journallog_blk
- 不支持remount pstore 分区之后依旧生成文件

注意事项

- 当前设计仅对shutdown生效，即调用call chain是传入reason为SYS_POWER_OFF，降低改动影响范围：

```
static void kernel_shutdown_prepare(enum system_states state)
{
    blocking_notifier_call_chain(&reboot_notifier_list,
        (state == SYSTEM_HALT) ? SYS_HALT : SYS_POWER_OFF, NULL);
    system_state = state;
    usermodehelper_disable();
    device_shutdown();
}
```

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公版测试结果

- KE/REBOOT不生成blk版本日志, pass
- shutdown命令能生成blk版本日志, pass
- poweroff命令能生成blk版本日志, pass
- dump 整体耗时: 5ms

```
122.292584][T500001] systemd-shutdown: systemd-shutdown[1]: Rebooting.
122.292590][T500001] systemd-shutdown: pstore: [name:pstore&]pstore_ram_blk_notifier_call: reason 1, don't do dump!
122.292603][T500001] systemd-shutdown: ramoops: [name:ramoops&]pstore_ram_blk_open: open /dev/sdc18 suc, total size = 139460608!
122.293113][T500001] systemd-shutdown: ramoops: [name:ramoops&]pstore_ram_blk_copy_prz_buffer: suc dump prz to buffer!
122.296809][T500001] systemd-shutdown: ramoops: [name:ramoops&]pstore_ram_blk_dump: CONSOLE want write 262156 bytes, ret 262156 to ram_blk /dev/sdc18, pos 262156!
122.296951][T500001] systemd-shutdown: ramoops: [name:ramoops&]pstore_ram_blk_copy_prz_buffer: suc dump prz to buffer!
122.297635][T500001] systemd-shutdown: ramoops: [name:ramoops&]pstore_ram_blk_dump: PMSG want write 65548 bytes, ret 65548 to ram_blk /dev/sdc18, pos 327704!
122.297651][T500001] systemd-shutdown: pstore: [name:pstore&]sdl_debug: dump blk log time 5(ms)
```

- Read 整体耗时:11ms

```
l"rw-rw---- 1 root root    19 Jul  2 04:44 vcp_log_17__2025_0702_044426          oot@auto8676p164hyp:boot__normal# cat kernel_l
<6>[ 0.154890][T400001] swapper/0: ramoops: [name:ramoops&]sdl_debug: ramoops_register_dummy do ram_blkdev path /dev/sdc18
<6>[ 0.156335][T400001] swapper/0: ramoops: [name:ramoops&]sdl_debug: ramoops_register_dummy do ramoops_max_reason = 2
<6>[ 0.177646][T1400001] systemd: [name:pstore&]sdl_debug: get blk log time 11(ms)
oot@auto8676p164hyp:boot__normal#
```

Thank you
Questions and Discussions